

Design-Document : Trupal Jageshkumar's Homepage

1. Project description

A static, front-end-only personal homepage for a computer-vision & machine-learning engineer. Its job is to introduce the person, show credible proof of work, and make it effortless to get in touch — while demonstrating clean, standards-based HTML, CSS, and vanilla ES6.

The visual concept is a **plotter's notebook**: warm graph-paper, heavy grotesque poster type, and a navy + magenta + cobalt palette. The hero is a **live regression playground** — click to drop data points and a least-squares line refits in real time. It is a literal nod to the subject matter (fitting a model to noisy data) and gives the page a memorable identity without stock imagery.

Scope: four pages (Home, Projects, About, Blog), no backend, no framework, no component library. Home, Projects, and About are written by the author; the Blog article is the AI-generated page. Deployed as a public static site.

Non-goals: blog CMS, contact form backend, authentication, analytics.

2. User personas

Persona A — Mansi, the recruiter

- **Age / role:** 25, technical recruiter at a mid-size company.
- **Context:** skims dozens of portfolios a day, often on a laptop between calls.
- **Goals:** quickly judge whether this person's skills fit an open ML role, and find a way to contact them.
- **Frustrations:** portfolios that bury the person's actual skills under visual noise, or hide the contact details.
- **Needs from the page:** a one-line summary of who this is, a scannable list of focus areas, and an obvious contact action.

Persona B — Margi, the hiring engineer

- **Age / role:** 28, senior ML engineer who will interview the candidate.
- **Context:** wants substance — did this person actually ship anything?
- **Goals:** read a couple of project write-ups and check that the work is real.
- **Frustrations:** vague project cards with no outcome or decision explained.
- **Needs from the page:** project pages with the problem, the approach, and the result stated plainly; links to code.

Persona C — Janvi, a fellow student / peer

- **Age / role:** 23, classmate exploring how others built their homepage.
 - **Context:** curious about the tech and the “creative addition.”
 - **Goals:** see the interactive component and understand the build.
 - **Frustrations:** sites that are impressive but impossible to learn from.
 - **Needs from the page:** a working, delightful demo and a clear README/repo.
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3. User stories

These are the use cases, written as short stories.

Story 1: First Impression

A recruiter finds my resume link and clicks through. She lands on my homepage and, within a few seconds, sees my name and that I'm a CS grad student at Northeastern building computer-vision and machine-learning systems; nothing else is competing for her

attention. If she wants more, the nav bar is right there, and my email is one click to copy.

Story 2: Going Deeper

A hiring engineer wants to know whether my research and projects are real, so he opens the Projects page. Instead of a wall of text for every project, he sees one line summarising each, and only expands the ones he's curious about — the Jetson Nano detection work published at ICT4SD, the IoT waste-bin monitor, or the satellite image processing I did at ISRO — each linking to its paper. The longer write-ups live on their own page, so Home stays short for people who don't want them.

Story 3: Trying It Out

A fellow student is curious about the creative part of the site. On Home he clicks the regression plot to drop points and watches the best-fit line update live, then drags a satellite into orbit in the little gravity sandbox. Because he's a keyboard person, he presses ⌘K and moves around the site without touching the mouse — and a small hint tells him the shortcut is there, so he isn't expected to guess it.

Story 4: Reaching Out

Someone lands on the site just wanting my email and GitHub, and finds them without hunting — a copy-email button and clear links. Because it's a static site, the half-minute they give it is spent reading, not watching a spinner.

Story 5: For Everyone

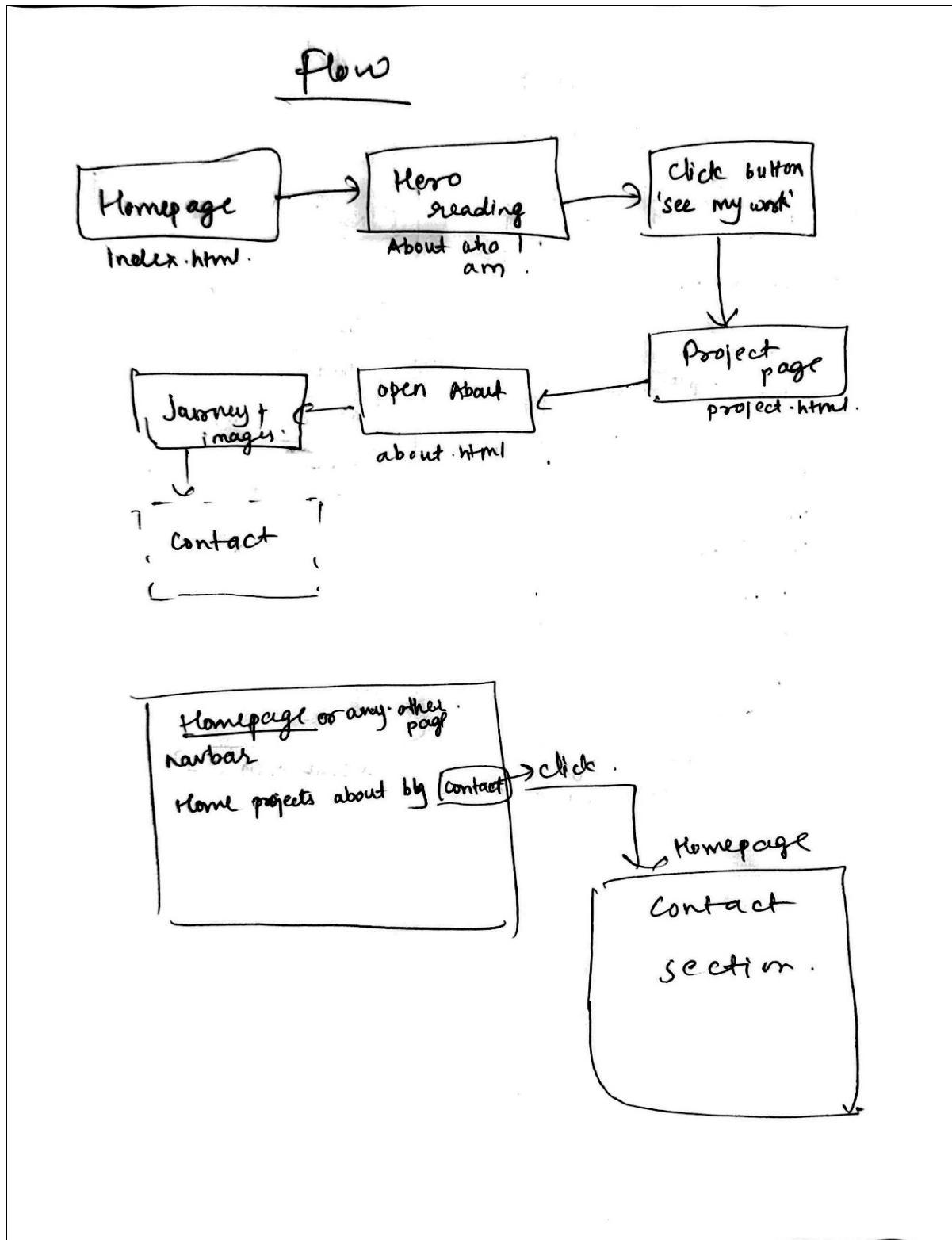
A screen-reader user can use the whole site, because every image is described and every control is a real button or link. Someone who prefers a darker screen or less motion gets a dark theme and reduced-motion support that the site remembers.

4. Design tokens

- **Palette (light, default):** `--paper #f6f4ee` (warm graph paper), `--ink #17203a` (near-black navy), `--hot #ff2e7e` (magenta, data points & highlights), `--cool #1f5bd6` (cobalt, the fitted line & links).
- **Palette (dark):** `--paper #0f1320`, `--ink #f1f3f8`, `--hot #ff5b98`, `--cool #79a6ff`.
- **Type:** Bricolage Grotesque (characterful display), Hanken Grotesk (highly legible body), Space Mono (data labels, equations, nav — only where it is genuinely data).
- **Accent strategy:** two accents with distinct jobs — magenta = data, cobalt = the model. Deliberately avoids the cream-and-terracotta and single-acid-accent-on-black defaults.
- **Signature devices:** a graph-paper background grid, hard 1.5px ink borders, and offset drop-shadows (a plotted, printed feel rather than soft SaaS cards).

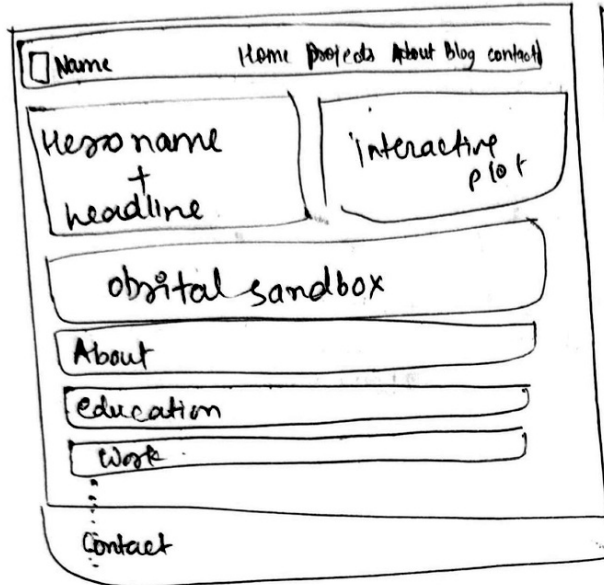
5. Design mockups (wireframes)

I sketched each page by hand to work out the layout — what goes where, how the navigation connects the pages, and roughly what I'll place in each section. These are my original hand-drawn wireframes.

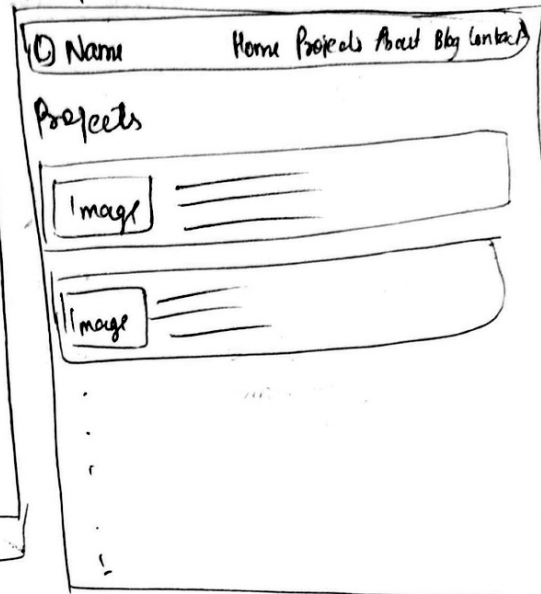


Page layouts (desktop)

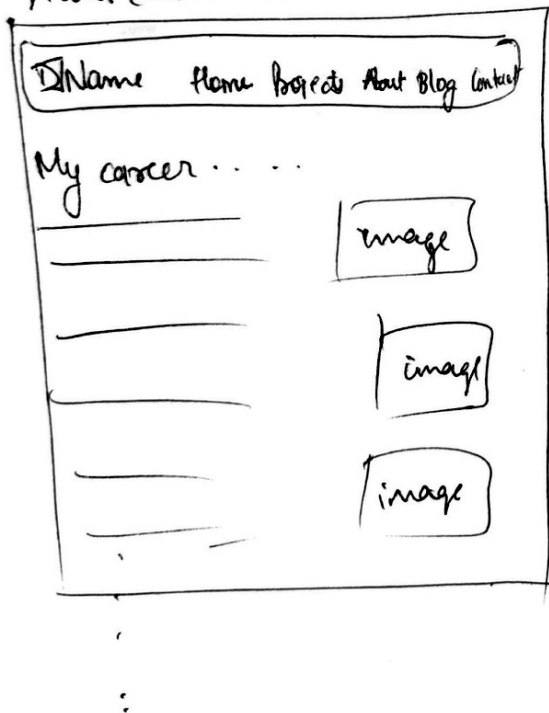
Home (index.html)



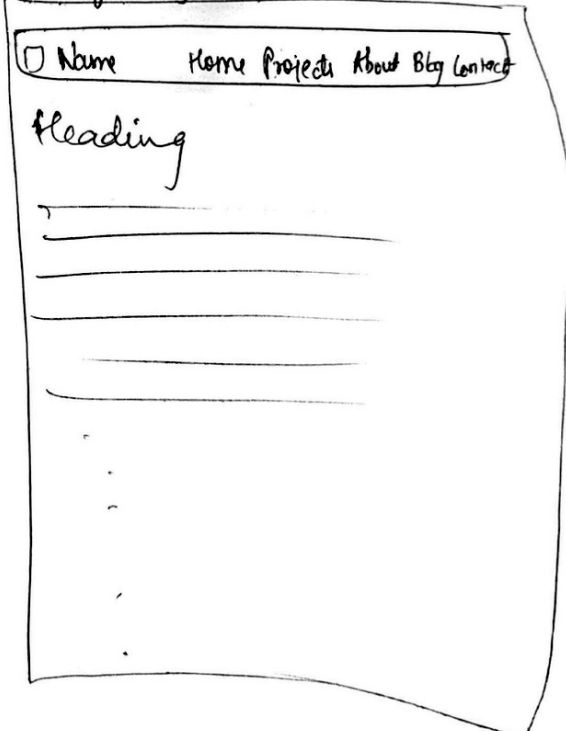
Projects (projects.html)

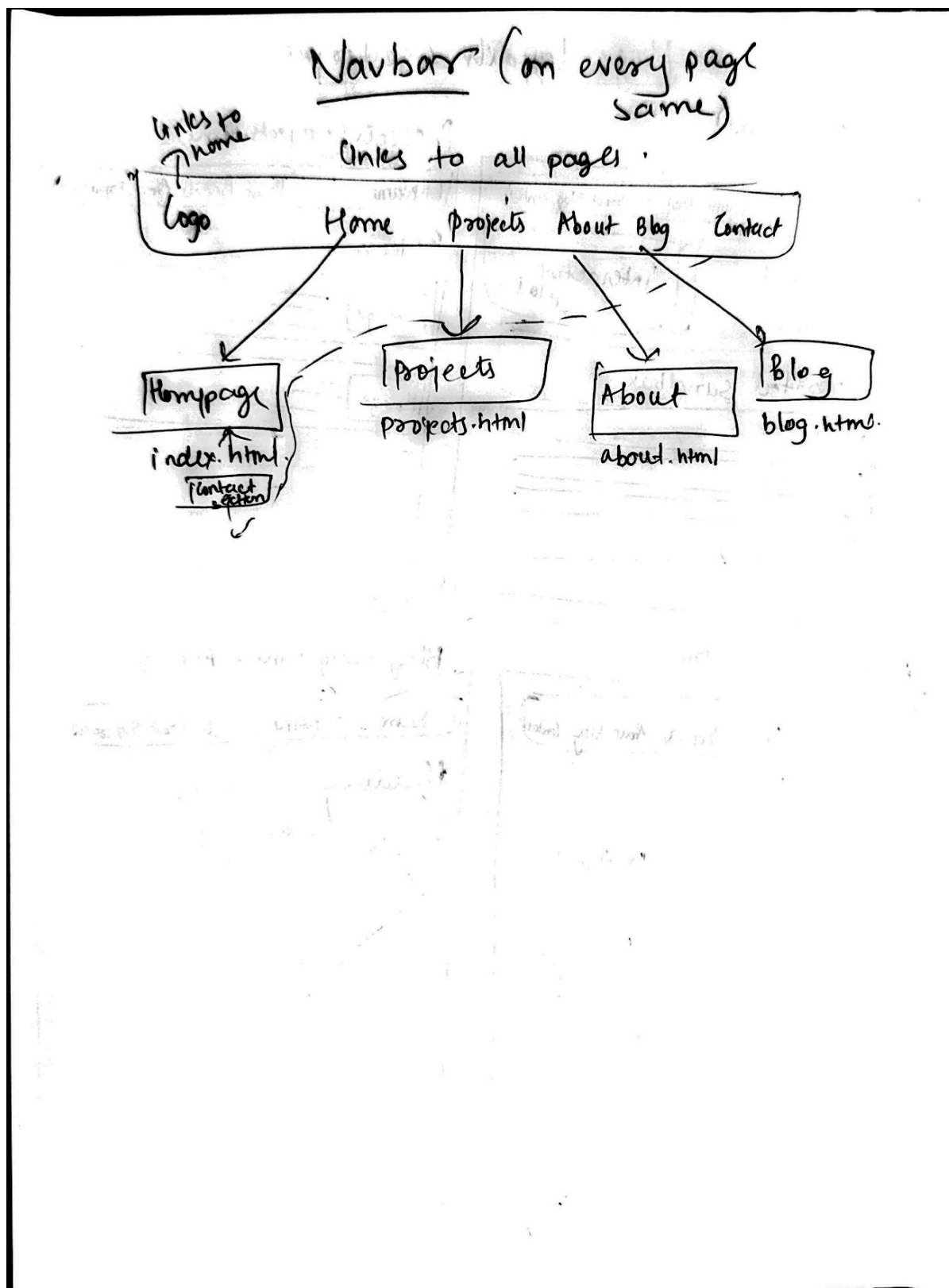


About (about.html)

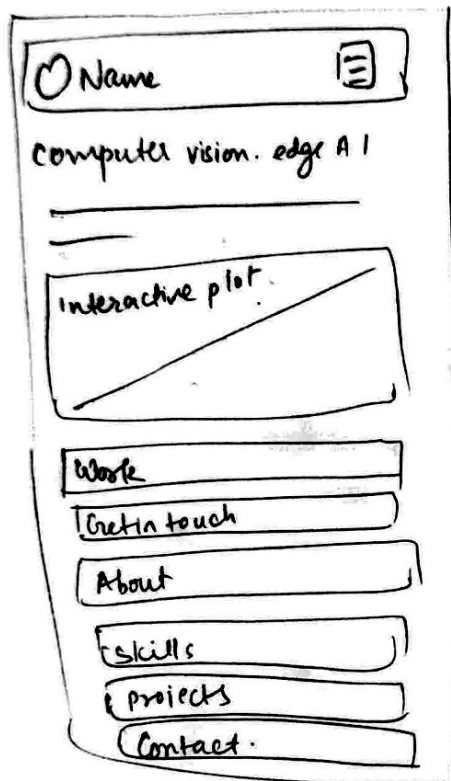


Blog (blog.html) - At page

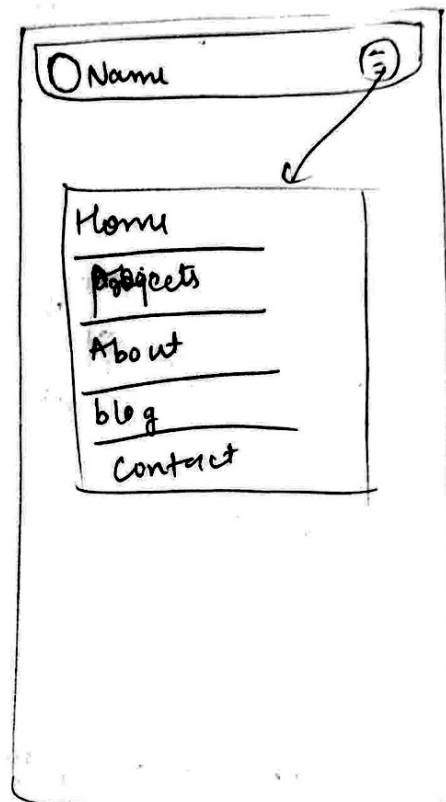




Mobile layout



Homepage
(Mobile)



Menu open (≡)

6. Design rationale (self-critique)

- **One bold element, quiet everywhere else:** the regression playground is the single memorable moment; every other section is calm, bordered, and consistent so the hero stands out.
- **Structure encodes meaning:** section labels are written as function calls (`about()`, `focus()`, `projects()`) — an on-theme affordance for a data page, not generic 01 / 02 / 03 numbering, since the sections are not a sequence.
- **Motion is purposeful:** there is no ambient animation at all; the only motion is the plot responding to your clicks (and button hovers). This avoids the scattered scroll-reveal effects that read as generated, and it means `prefers-reduced-motion` users lose nothing.
- **Chanel's rule:** cut anything that only decorates. The graph paper, borders, and offset shadows each signal “plotted/printed”; the palette stays to three inks plus paper.